



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

LAB ASSIGNMENT-2

Submitted by:

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1. Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.

Data Members (properties):

Name

Roll No

Degree

Hostel

Current CGPA

Member Functions (behavior):

addDetails()

updateDetails()

updateCGPA()

updateHostel()

displayDetails()

```
1  #include<iostream>
2  using namespace std;
3  struct student{
4      char Name[20];
5      int rollno;
6      char hostel[5];
7      float CGPA;
8      char degree[10];
9      void addDetails(){
10         cout<<"Enter name:";
11         cin>>Name;
12         cout<<"Enter rollno:";
13         cin>>rollno;
14         cout<<"Enter degree:";
15         cin>>degree;
16         cout<<"Enter hostel:";
17         cin>>hostel;
18         cout<<"Enter CGPA:";
19         cin>>CGPA;
20     }
21     void updateDetails(){ (const char [
22         cout<<"Enter updated name:";
23         cin>>Name;
24         cout<<"Enter updated rollno:";
25         cin>>rollno;
26         cout<<"Enter updated degree:";
27         cin>>degree;
28     }
29     void updatehostel(){
30         cout<<"Enter updated hostel:";
31         cin>>hostel;
32     }
33     void updateCGPA(){
34         cout<<"Enter updated CGPA:";
35         cin>>CGPA;
36     }
37     void displaydetails(){
38         cout<<"\nName:"<<Name;
39         cout<<"\nRollno:"<<rollno;
40         cout<<"\nDegree:"<<degree;
41         cout<<"\nHostel:"<<hostel;
42         cout<<"\nCGPA:"<<CGPA;
43     }
44 };
45
```

```

45
46     int main(){
47         student s1;
48         s1.addDetails();
49         s1.updateDetails();
50         s1.updatehostel();
51         s1.updateCGPA();
52         s1.displaydetails();
53         return 0;
54     }
55

```

OUTPUT:

```

● Enter name:Ryan
Enter rollno:1024150188
Enter degree:ENC
Enter hostel:D
Enter CGPA:8.96
Enter updated name:Shivam
Enter updated rollno:1024150189
Enter updated degree:ENC
Enter updated hostel:L
Enter updated CGPA:

```

2. Differentiate between private and public access / scope.

Perform Question No. 1 using class instead of structure, having the data members private and some member functions in private scope and some in public scope

OOPS > A2 > Q2.cpp > ...

```
1  #include<iostream>
2  using namespace std;
3  class student{
4      private:
5          char Name[20];
6          int rollno;
7          char hostel[5];
8          float CGPA;
9          char degree[10];
10     void addDetails(){
11         cout<<"Enter name:";
12         cin>>Name;
13         cout<<"Enter rollno:";
14         cin>>rollno;
15         cout<<"Enter degree:";
16         cin>>degree;
17         cout<<"Enter hostel:";
18         cin>>hostel;
19         cout<<"Enter CGPA:";
20         cin>>CGPA;
21     }
22     public:
23     void pubaddDetails(){
24         addDetails();
25     }
26     void updateDetails(){
27         cout<<"Enter updated name:";
28         cin>>Name;
29         cout<<"Enter updated rollno:";
30         cin>>rollno;
31         cout<<"Enter updated degree:";
32         cin>>degree;
33     }
34     void updatehostel(){
35         cout<<"Enter updated hostel:";
36         cin>>hostel;
37     }
38     void updateCGPA(){
39         cout<<"Enter updated CGPA:";
40         cin>>CGPA;
41     }
```

```

42         }
43         void displaydetails(){
44             cout<<"\nName:"<<Name;
45             cout<<"\nRollno:"<<rollno;
46             cout<<"\nDegree:"<<degree;
47             cout<<"\nHostel:"<<hostel;
48             cout<<"\nCGPA:"<<CGPA;
49         }
50     int main(){
51         student s1;
52         s1.pubaddDetails();
53         s1.updateDetails();
54         s1.updatehostel();
55         s1.updateCGPA();
56         s1.displaydetails();
57         return 0;
58     }
59

```

OUTPUT:

```

PS C:\codes\UNC401> cd "c:\codes
● Enter name:Ryan
Enter rollno:1024150188
Enter degree:ENC
Enter hostel:D
Enter CGPA:9
Enter updated name:Hardik
Enter updated rollno:1024150365
Enter updated degree:ENC
Enter updated hostel:D
Enter updated CGPA:

8

Name:Hardik
Rollno:1024150365
Degree:ENC
Hostel:D
CGPA:8
○ PS C:\codes\UNC401\OOPS\A2>

```

3.Create a code snippet that illustrates the following:
Calling of private member functions inside public member function.

```

OOPS > A2 > Q3.cpp > main()
1  #include<iostream>
2  using namespace std;
3  class call_priv{
4  private:
5      void privateFunc(){
6          cout<<"Private function called\n";
7      }
8  public:
9      void publicFunc(){
10         privateFunc();
11     }
12 };
13 int main(){
14     call_priv d;
15     d.publicFunc();
16     return 0;
17 }
18

```

OUTPUT:

```
PS C:\codes\UNC401> cd c:\c
Private function called
PS C:\codes\UNC401\OOPS\A2>
```

4. Define a class **Rectangle** with variables **width** and **height** of integer type along with the following:

- (a) void `getdata()` to initialize object values
- (b) void `calculateArea()` to calculate and display the area

```
OOPS > A2 > G+ Q4.cpp > main()
1  #include<iostream>
2  using namespace std;
3  class Rectangle{
4      int width,height;
5      public:
6          void getdata(){
7              cout<<"Enter width:";
8              cin>>width;
9              cout<<"Enter height:";
10             cin>>height;
11         }
12         void calculateArea(){
13             cout<<"Area of Rectangle:"<<width*height<<endl;
14         }
15     };
16     int main(){
17         Rectangle r;
18         r.getdata();
19         r.calculateArea();
20         return 0;
21     }
22
```

OUTPUT:

```

● PS C:\codes\UNC401\OOPS\A2> cd "c:\
Enter width:55
Enter height:33
Area of Rectangle:1815
○ PS C:\codes\UNC401\OOPS\A2>

```

5. Define a class **Complex** with variables **real** and **imaginary** along with the following:

- (a) void setComplex(float, float) to initialize object values
- (b) void displayComplex() to show the complex number
- (c) Pass and return objects to calculate sum of two complex numbers. Display the sum.

```

OOPS > A2 > Q5.cpp > complex > displayComplex()
1  #include<iostream>
2  using namespace std;
3  class complex{
4      public:
5          float real;
6          float imag;
7          void setComplex(){
8              cout<<"Enter real part value: ";
9              cin>>real;
10             cout<<"Enter imaginary part value: ";
11             cin>>imag;
12         }
13         void displayComplex(){
14             if(imag>=0)
15                 cout<<real<<"+"<<imag<<"i\n";
16             else
17                 cout<<real<<imag<<"i\n";
18         }
19         complex addComplex(complex c){
20             complex temp;
21             temp.real=real+c.real;
22             temp.imag=imag+c.imag;
23             return temp;
24         }
25     };
26     int main(){

```



```

26  int main(){
27      complex c1,c2,sum;
28      cout<<"Enter first complex number\n";
29      c1.setComplex();
30      cout<<"Enter second complex number\n";
31      c2.setComplex();
32      sum=c1.addComplex(c2);
33      cout<<"\nFirst Complex Number: ";
34      c1.displayComplex();
35      cout<<"Second Complex Number: ";
36      c2.displayComplex();
37      cout<<"Sum of Complex Numbers: ";
38      sum.displayComplex();
39      return 0;
40  }

```

OUTPUT:

```

● Enter first complex number
  Enter real part value: 5
  Enter imaginary part value: 2
  Enter second complex number
  Enter real part value: 4
  Enter imaginary part value: 6

  First Complex Number: 5+2i
  Second Complex Number: 4+6i
  Sum of Complex Numbers: 9+8i
○ PS C:\codes\UNC401\OOPS\A2>

```

6.Implement scope resolution operator :: for the following uses:

- Class functions defined outside the class
- Access a global variable with the same name as a local variable
- Access static variables
- Use inbuilt libraries (cin, cout) with scope resolution operator

```

1  #include<iostream>
2  int x=100;
3  class Demo{
4  public:
5      static int value;
6      void show();
7  };
8  int Demo::value=50;
9  void Demo::show(){
10     std::cout<<"Function defined outside class using scope resolution\n";
11 }
12 int main(){
13     Demo d;
14     d.show();
15     int x=20;
16     std::cout<<"Local x = "<<x<<std::endl;
17     std::cout<<"Global x = "<<::x<<std::endl;
18     std::cout<<"Static class variable = "<<Demo::value<<std::endl;
19     int num;
20     std::cout<<"Enter a number: ";
21     std::cin>>num;
22     std::cout<<"Number entered = "<<num<<std::endl;
23     return 0;
24 }
25

```

OUTPUT:

```

PS C:\codes\UNC401\OOPS\A2> cd "
Function defined outside class u
Local x = 20
Global x = 100
Static class variable = 50
Enter a number: 

```

7. Implement *namespace* in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

```

1  #include<iostream>
2  using namespace std;
3  namespace First{
4      int num=10;
5      void display(){
6          cout<<"First namespace number = "<<num<<endl;
7      }
8  }
9  namespace Second{
10     int num=20;
11     void display(){
12         cout<<"Second namespace num = "<<num<<endl;
13     }
14 }
15 int main(){
16     cout<<First::num<<endl;
17     cout<<Second::num<<endl;
18     First::display();
19     Second::display();
20     return 0;
21 }
22

```

OUTPUT:

```

● 10
  20
  First namespace number = 10
  Second namespace num = 20
○ PS C:\codes\UNC401\OOPS\A2>

```